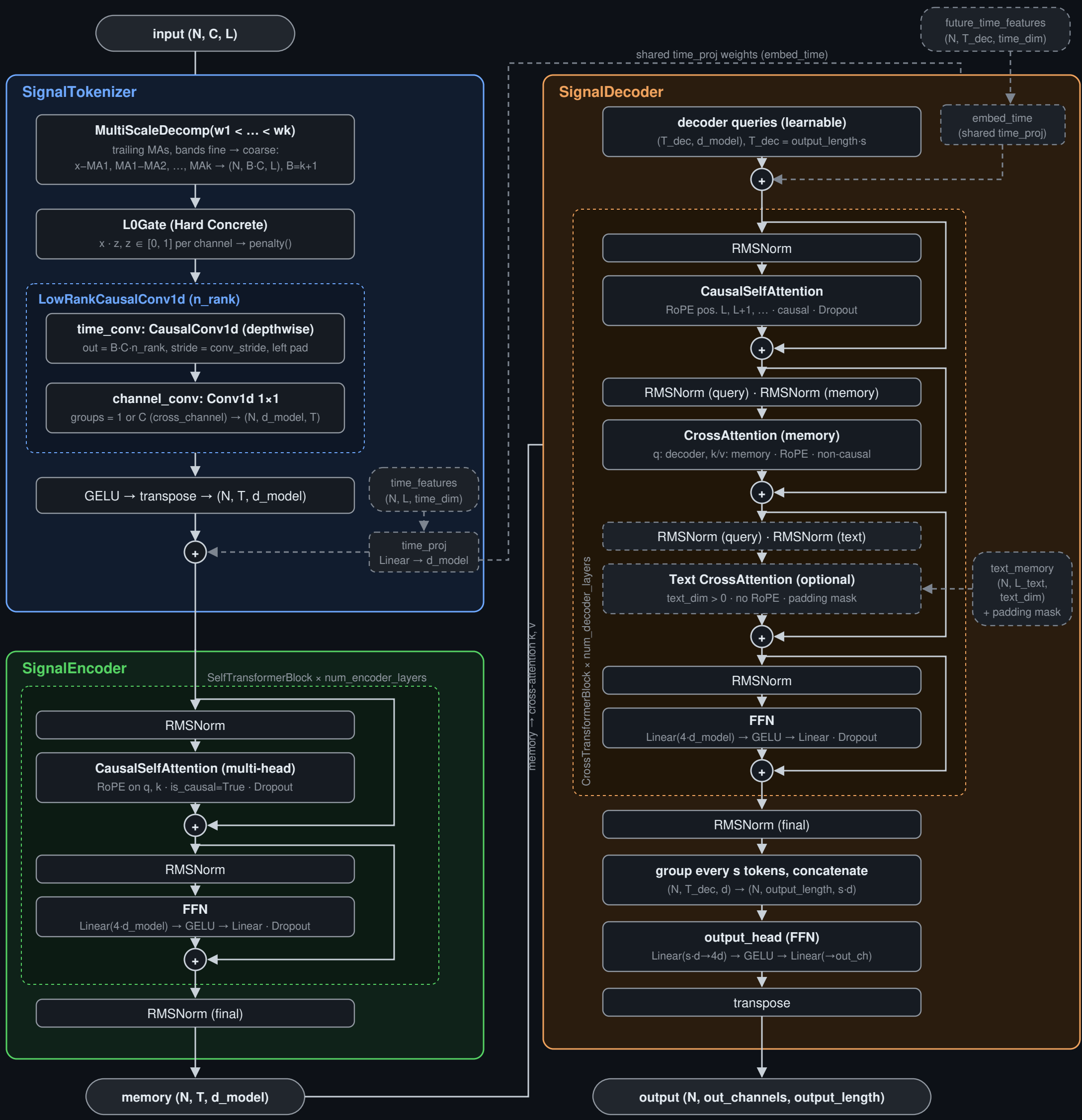
Signalformer architecture

multi-scale causal decomposition + convolutional tokenizer + causal encoder-decoder Transformer



Legend

tensor flow — optional input (time_dim > 0, text_dim > 0) — ⊕ addition / residual connection

Causality: every token at time t depends only on inputs at times ≤ t (left-padded convs, trailing MAs, causal masks).
Decoder cross-attention sees all of memory, which is entirely past data. N batch, C signals, L input length, B bands, T = ceil(L / conv_stride), s = samples_per_output.